The empirical recurrence for OEIS A206088, proved by transfer matrix

Adrian Perez Fontelles
Independent researcher

2 September 2026

Abstract

OEIS A206088 counts arrays over $\{0,\dots,2\}$ under a condition on the perimeter cycle of every 2×3 and 3×2 subblock, and carries an empirical recurrence of order 4 contributed by R. H. Hardin. It is true, and it is decidable rather than empirical. The condition is local to a bounded window of consecutive rows, so the count is a walk count on $S = 270$ states and is C-finite; whether the conjectured recurrence annihilates it is then settled by a finite exact computation, with the Cayley–Hamilton theorem bounding the work.

2020 Mathematics Subject Classification. 05A15, 05B45, 15B36.

1 The conjecture

OEIS A206088 is “Number of (n+1) X 3 0..2 arrays with every 2 X 3 or 3 X 2 subblock having exactly two counterclockwise and two clockwise edge increases.”. It has offset 1 and begins

$$270, 780, 2016, 5952, 17976, 54432, 165936, 504912, \ \dots$$

The entry states, as a conjecture contributed by its author and never marked settled:

> Empirical: a(n) = a(n-1) + 6*a(n-2) + 4*a(n-3) - 10*a(n-4) for n>6.

As of the “Last modified” line on the live entry (Jun 13 2018 11:23:42, revision 12) this is still recorded as empirical, and nothing on the entry records it as proved.

2 What is being counted

Both a 3×2 and a 2×3 window have all six of their cells on the perimeter, so each carries a 6-cycle. For $\begin{pmatrix} p & q \\ r & s \\ t & u \end{pmatrix}$ the clockwise cycle is $p \to q \to s \to u \to t \to r \to p$; for $\begin{pmatrix} p & q & t \\ r & s & u \end{pmatrix}$ it is $p \to q \to t \to u \to s \to r \to p$. An edge increase is a step of the cycle that goes up in value, and counterclockwise counts the same cycle traversed the other way. The entry requires of every window of either shape that it has exactly 2 counterclockwise and 2 clockwise edge increases.

The readings were pinned by direct enumeration before anything was built: over $\{0,1,2\}$ the 3×2 arrays of clockwise count 2 number 423 and those of count 3 number 150; over $\{0,\dots,3\}$ the 3×2 arrays of count 1 number 480; and the 2×3 arrays over $\{0,1,2\}$ of clockwise count 2 also number 423. Those are the first terms the corresponding entries publish.

3 The count is a walk count

A 3×2 window spans three consecutive array rows and a 2×3 window two, so take as state the PAIR of consecutive rows. The 2×3 windows lie inside a single pair, so they decide which pairs are admissible at all; the 3×2 windows straddle three rows, so they decide which pair may follow which. A step appends one array row, turning (r, s) into (s, t) and testing every 3×2 window of r, s, t at once. An $(n + 1)$-row array is a walk of $n - 1$ steps and

$$a(n) \;=\; \iota^\top M^{n-1} \tau, \qquad \iota = \tau = (1, \ldots, 1)^\top,$$

on the $S = 270$ admissible pairs.

At $n = 1$ the array has two rows, so no 3×2 window fits and only the 2×3 windows apply; the entry's first term is 270, which the model reproduces along with every later term. In particular a is C-finite. The alignment of the walk index with the entry's own n was checked against its published terms rather than assumed.

4 The proof

Lemma 1. *Let $q(t) = t^4 - \sum_i c_i t^{4-i}$ be the characteristic polynomial of the conjectured recurrence. The residual $a(n) - \sum_i c_i a(n-i)$ equals $\iota^\top M^j q(M) \tau$ for the corresponding walk index j, so the recurrence holds from some point on if and only if $\iota^\top M^j q(M) \tau = 0$ for all large j, and it is enough to examine $j < S$.*

Proof. Factor M^j out of $M^{j+4} - \sum_i c_i M^{j+4-i}$. For the bound, M^S is an integer combination of $I, M, \ldots, M^{S-1}$ by Cayley–Hamilton, so the numbers $u_j = \iota^\top M^j q(M) \tau$ obey the monic recurrence given by the characteristic polynomial of M; S consecutive zeros force every later one, and the last nonzero u_j pins the threshold exactly. □

Theorem 2.

$$a(n) \;=\; a(n-1) + 6\,a(n-2) + 4\,a(n-3) - 10\,a(n-4)$$

for every $n > 6$.

Proof. The vector $w = q(M)\tau$ was formed by 4 matrix–vector products in exact integer arithmetic and $\iota^\top M^j w$ evaluated for $j = 0, 1, \ldots,$ again exactly; those integers vanish from the index corresponding to $n = 6 + 1$ onwards and the run continued until the working vector was identically zero, which settles every larger j at once. The bound is exact: at $n = 6$ the recurrence fails on the entry's own published terms, so no larger range holds. □

5 Verification

Three checks, each independent of the algebra above.

First, the model reproduces all 22 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry: the digraph is built by reading the entry's English, and a misreading gives different counts.

Second, the conjectured recurrence was evaluated directly on those published terms, with no matrices involved, and holds wherever the proved range and the data overlap.

Third, the annihilation test of Lemma 1 was carried out in exact integer arithmetic on the whole vector rather than on a sampled prefix.

References

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